# Human Action Recognition Methodologies and its State of the art

Karthik Uppuluri
Rice University
6100 Main St, Houston, TX-77251
Karthik.Uppuluri@rice.edu

## Abstract

*Action recognition is a process of labeling an action instance with an action verb which best describes the given instance. Many applications of action recognition include video-editing, motion capture, automatic annotation of videos, video retrieval and surveillance. In this paper I discussed the basic work-flow of an action recognition system and different methodologies in building an action recognition system. And I also discussed the present state-of-the-art algorithms in action recognition.*

***Keywords**: Human action recognition, action recognition methodologies*

## 1. Introduction and Motivation

Action recognition is a process of recognizing the "actions and goals" of one or more objects by analyzing a series of observations on the objects using a computer system [1]. For instance, a typical objective of an action recognition system present in a car could be to recognize the movement of pedestrians and animals on the road or understand the state of the driver to prevent him from sleeping etc. Similarly a goal of a poacher could be to harm an animal; action recognition systems can help recognizing such violent intentions.

Leonardo Davinci is the first one to study about the human body anatomy and representations [2][3]. Giovanni Alfonso Borelli studied about the biomechanics of the body [4]. His studies included muscle analysis and mathematics of movements like running, jumping etc. Etienne-Jules Marey and Eadweard Muybridge are the pioneers in the field of motion pictures [5]. Gunnar Johnson studied the usage of image sequences for programmed human motion analysis. He used multiple LED based light displays and image sequences to analyze human motion. Max Wertheimer's [6] study on perceived motion and figural organization formed the basis for present day motion capture

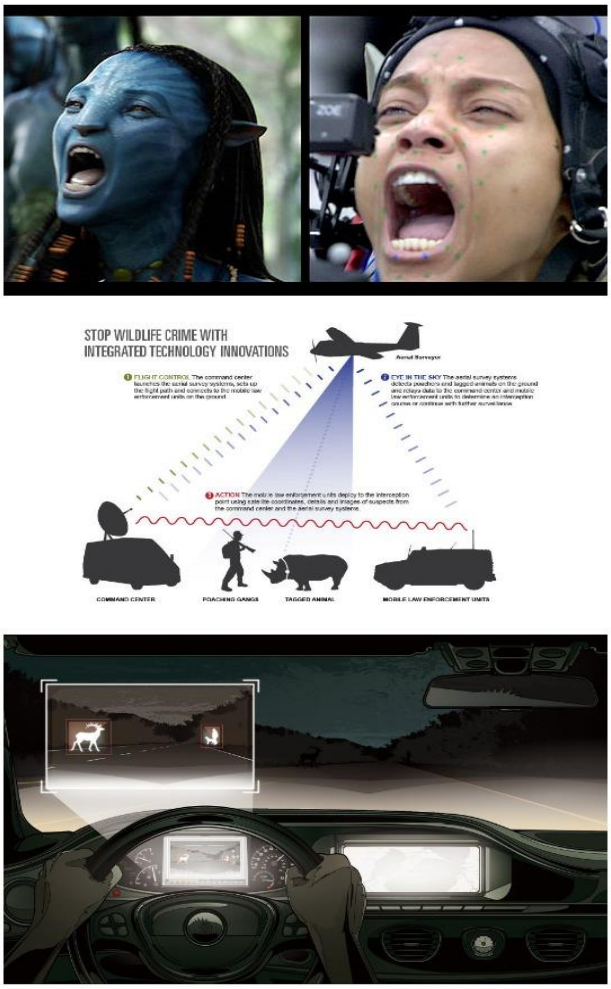


Figure.1 Applications of Action Recognition Row1 Motion capture [7], Row2 Intelligent Wildlife Surveillance [8], Row3 Driver Assistance [9]

There are lots of applications of action recognition in today's world. Action recognition systems are used in automatic annotation of videos, human-computer interaction (Kinect), video editing etc. Given below are three most interesting and motivating applications of action recognition

**Applications of Action recognition**

a. **Intelligent driver Assistance**

Driver assistance is one of widely used applications of Action recognition. Action recognition systems can help the driver in

1. Preventing Pedestrian/animal collision by understanding the pedestrian and animal movement.
2. Drowsiness detection by tracking the movements of the eyes and head
3. Providing pre-crash warnings by tracking other vehicles and obstacles approaching

**b. Intelligent Wildlife surveillance**

Poaching and deforestation are two major problems faced by today's wildlife. Action recognition systems play a major role in preventing both. World Wildlife Fund for Nature in collaboration with University of Zurich are creating Unmanned aerial vehicles which can be used to sense these two scenarios and report to the nearest Wildlife surveillance center.

**c. Motion Capture**

Motion capture is one of the most interesting applications of action recognition. It forms the basis for most of the computer generated Imagery used in movies and video games. Most of the today's video games use motion capture technology to generate the animated characters of martial artists, football players, athletes etc. Many Hollywood movies like Avatar, The Mummy and The lord of the rings used motion capture technology to generate the characters so that they make the animated characters more realistic by capturing different features like facial expressions, gait etc.

## 2. Methods

**Goal**: Determine the action label which is usually an action verb that best describes an action instance [10].

An action recognition system can be divided into three levels [10].

**a. Data Input**

The input to the recognition system depends on the objective of the system. For instance if the objective of the system is to recognize a martial art movement, then the input is a set of action instances of different martial art movements.

**b. Feature extraction**

Feature extraction can be categorized into two levels [11]

1. Local Feature extraction
2. Global Feature extraction

**Local Feature extraction**

Local feature extraction is a process of extracting high interest regions (points) in the input video. There are there main kinds of local features extracted [11]

**Spatio-temporal Interest points**

These are the points of high gradient in the space-time volume of the video [11]. These points can be treated as 3D SIFT points. A typical SIFT feature is extracted by [12]

1. Computing a scale space approximation of the image
2. Constructing the Laplacian of Gaussian approximation for the scale space
3. Finding the key points which the local maxima in the scale space
4. Eliminate the edges and low contrast features
5. Assigning orientation to the key points found.

So the only difference is that the scale space is constructed with temporal and spatial scale parameters to extract Spatio-temporal features.

**Cuboidal Features**

Once a Spatio-temporal key point is found, the Spatio-temporal cuboids are extracted at each key point. The size of the cuboid is chosen in such a way

that it contains most of the information which made the key point unique. Once the cuboid is extracted it is normalized to make them invariant to changes in translation and appearance.

**Volumetric Features**

Optical flow of the video is used to extract the volumetric features. Optical flow consists of a vertical component and a horizontal component for each frame. There are two kinds of volumetric features extracted 'one box features' which is the cumulative sum of the optical flow component and 'two box features' which is the differences in the cumulative sum of combinations of the two boxes over the stack of n frames.

**Global Features** [11]

**Motion Descriptors**

A figure centric Spatio temporal volume is created by tracking the human figure, which in turn is done by constructing a window in each frame centered at the figure. Once the Spatio-temporal volume is constructed optical flow is computed at each frame. Optical flow consists of two components the horizontal component and a vertical component. Both horizontal and vertical components are split in four non-negative channels and are normalized by smoothing with a Gaussian kernel. Once the four channels are normalized the concatenation of these four channels gives a motion descriptor at each frame.

**Histogram of optical flow (HOF)**

Once the optical flow is computed a distribution of the direction of the flow gives the histogram of optical flow. Before the histogram is computed the magnitude of the flow vector is normalized to ensure scale invariance.

**Space-Time shapes**

For each frame of the video a silhouette is extracted by subtracting the background and the concatenation of all these silhouettes gives a space time shape for each video or the action instance.

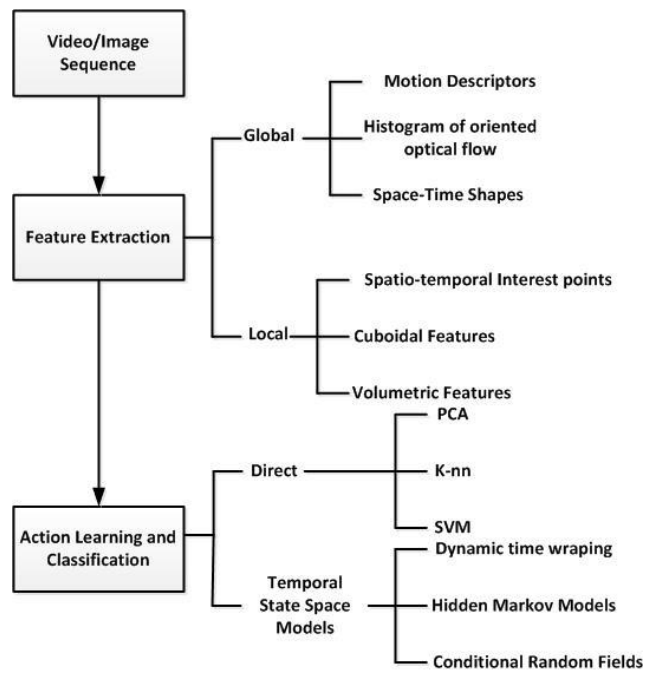


Figure2. Data Flow and methods in an action Recognition System

**c. Action learning and classification**

Once the features are extracted the next task is to learn from these features and create a classifier which can recognize these actions.

There are two kinds of the classification models widely used [13]

**Direct classification**

These classification algorithms do not consider the temporal domain; rather they combine all the frames of the action instance into a single representation or perform the analysis frame wise. Some of the direct classification algorithms include

a. **Principal Component Analysis** (Dimensionality reduction and classification)
b. **K-nearest neighbor classification**
c. Discriminative classifiers like **Support Vector Machines** and **Relevance Vector Machines**

SVM and K-nn are more effective direct classifiers for action recognition systems.

**Temporal state space models**

**Dynamic time warping** is a technique which can be used to measure the similarity between two temporal sequences which may vary in both speed and time [14]. The cost function used is called sequence alignment which is low if both frames are segmented similarly in time. Dynamic programming is used to calculate the optimal alignment.

**Hidden Markov Models** are another generative temporal state space models which can be used for action classification.
**Assumption**: System being modeled is a Markov Processes with unobserved states [15].
**Goal**: Recover the class label that is not immediately observable, but the other data that depends on the label is.

**Conditional Random Fields** are discriminative temporal state space models that can use multiple overlapping features to classify action instances. Once the classifier is learnt the action instances are detected by correlating these observed sequences to labeled sequences. Models like Bag-of-words with histogram similarity measurement can be used to do such correlation.

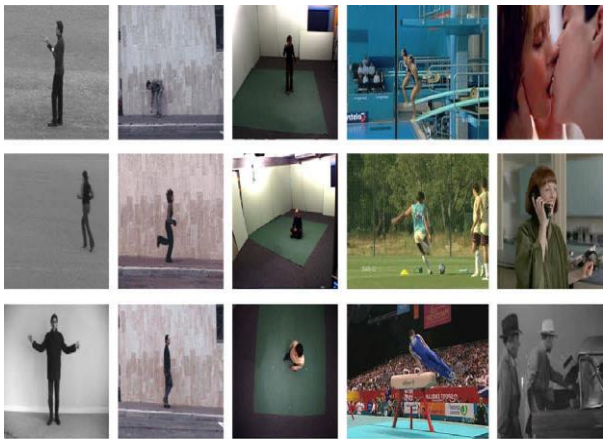

Figure.3 Datasets KTH (column1), Weizmann (coloumn2), INRIA (column 3), UCF (column 4), Hollywood (column5)[13]

**3. State-of-the-art**

There are five important data sets in the field of action recognition.

a. **KTH Human motion Dataset**[16]
   This dataset contains six actions namely walking, jogging running, boxing and Hand waving which are performed by 25 different actors under four different scenarios namely outdoors, outdoors with zooming, outdoors with different clothing and indoors.
b. **Weizmann Human Action Dataset**[17]
   This dataset contains 10 actions namely walk, run, jump, gallop sideways, bend, one-hand wave, two hands wave, jump in place, jumping jack and skip where each action is performed by 10 persons.
c. **INRIA Xmas multi-view Dataset**[18]
   This dataset contains 14 actions namely check watch, cross arms, scratch head, sit down, get up, turn around, walk, wave punch, kick, point, pick up, throw overhead and throw from bottom up performed by 11 persons and captured from five different viewpoints.
d. **UCF sports Dataset**[19]
   This dataset contains 150 sequences of sport-motions like diving, golf swinging, running, skating etc.
e. **Hollywood dataset**[20]
   This dataset contains eight actions namely answer phone, get out of car, handshake, hug, kiss, sit down, sit up and stand up performed by different actors.

The best accuracies are reported along with the features, feature descriptors and classifier used for KTH, UCF sports and Hollywood dataset in the table 1 shown below. For Weizmann dataset many research groups (Ikizler and Duygulu [21], Wang and Suter [22], Fathi and Mori [23], Schindler and van Gool [24], Tran and Sorokin [25], Weinland and Boyer [26] reported an accuracy of almost 100%. For INRIA Xmas dataset Liu and Shah [27] reported an accuracy of 82.8 % (2D) and Turaga and Veeraraghavan et.al [28] reported an accuracy of 98.8 % (3D).

| Dataset | Best Detector | Best Descriptor | Classifier | Accuracy | Research Group |
|---|---|---|---|---|---|
| KTH Human Motion | Harris 3D | Histogram of optical flow | Non-linear SVM | 92.1% | Wang et.al |
| UCF Sports | Dense | Histogram of gradients 3D | Non-linear SVM | 85.6 % | Wang et.al |
| Hollywood | Dense | Histogram of optical flow | Non-linear SVM | 47.4% | Wang et.al |

Table.1 Details of the best algorithms

## 4. Summary and Conclusion

In this paper I described the basic work flow of an action recognition system and different methodologies in building a system. I also provided the present state of the art of human action recognition systems. Usage of Histogram of optical flow as the feature descriptor and SVM as the classifier gave the best results. Present day action recognition systems are still challenged by various problems like occlusion, background clutter, different illuminations etc. [10]. More work is to be done on features, descriptors and classification algorithms which are insensitive to the problems mentioned above.

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